

Lecture 2

LIGHT

- Nature of Light-Newton's Corpuscular Theory-Huygens's wave Theory-Maxwell's electromagnetic Theory-Einstein's quantum Theory-Dual Nature theory

NATURE OF LIGHT

- The branch of physics that deals with the **production, emission and propagation of light** , its **nature and study** of the phenomena of **interference, diffraction and polarization** is called **physical optics**
- The basic principles regarding the nature of light were formulated in the latter half of the seventeenth century
- Until about this time, the general belief was that light consisted of a stream of particles called corpuscles

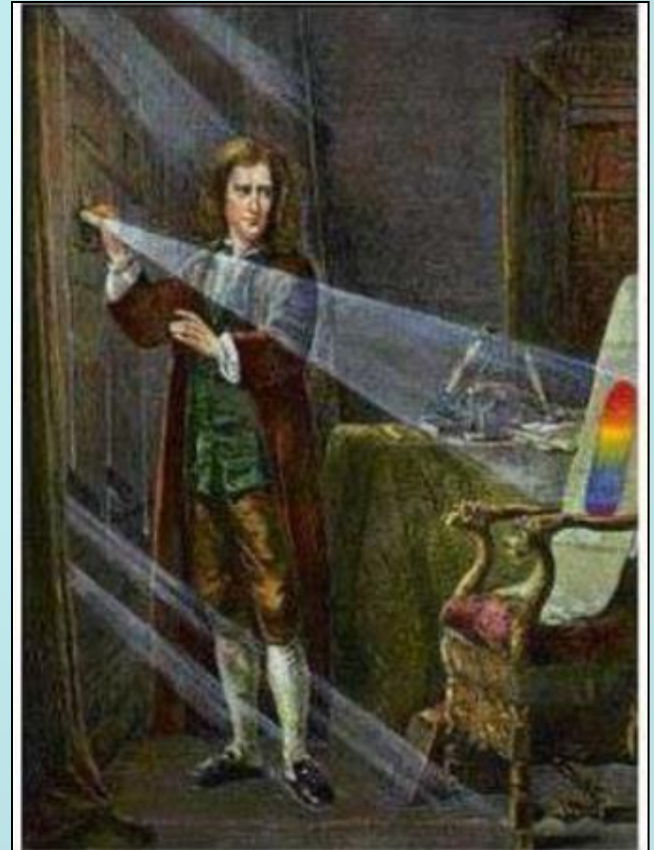
FOUR IMPORTANT THEORIES

Various theories have been put forward about the nature of light. We will make a brief summary of the four important theories which guided the evolution of our understanding of the nature of light

- **Corpuscular theory**
- **Wave theory**
- **Electromagnetic theory**
- **Quantum theory**

Newton's corpuscular theory

- The corpuscular theory of light was proposed by **Newton** in seventeenth century
- The main features of Newton's corpuscular theory of light are as follows:

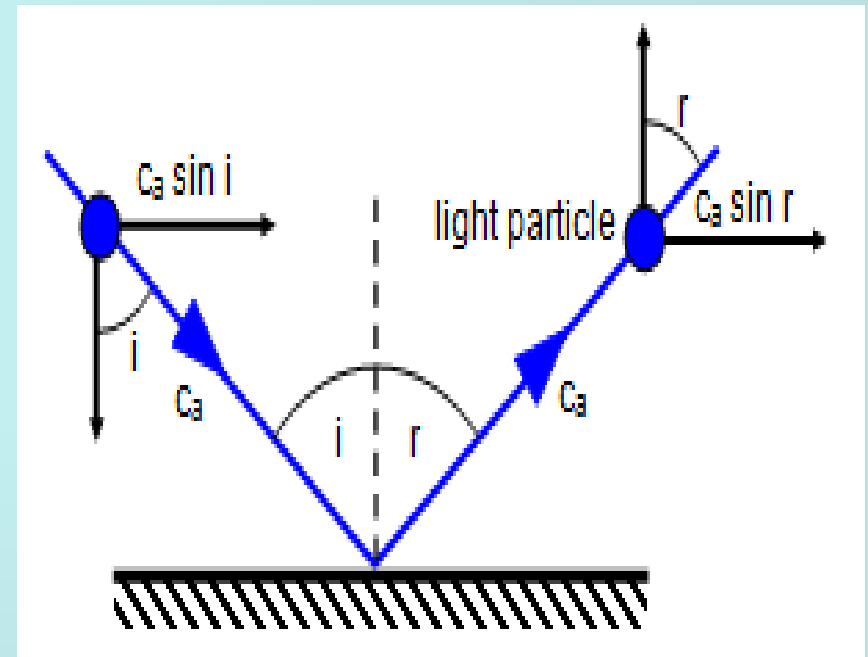


Features - corpuscular theory

- According to this theory, a luminous body continuously emits tiny, light and elastic particles called corpuscles.
- These particles or the corpuscles moves with high velocity as that of light and gets scattered in all directions of light.
- When these particles fall on the eye, they produce the sensation of vision
- This theory says that the velocity of light changes with the change in density of the medium in which it is used.
- This theory could explain three main phenomena of light that is reflection, refraction, and rectilinear propagation of light.
- This theory also says that corpuscles of different colors have different sizes.

Reflection of light on corpuscular theory

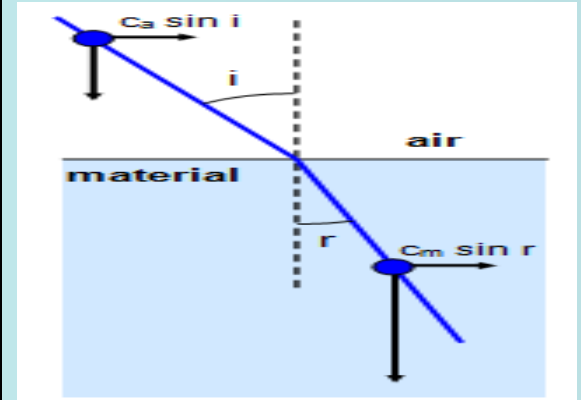
- Consider a particle of light in collision with a mirror.
- The collision is supposed to be perfectly elastic, and so the component of velocity perpendicular to the mirror is reversed while that parallel to the mirror remains unaltered.



- Component of velocity before collision parallel to the mirror = $c_a \sin i$
Component of velocity after collision parallel to the mirror = $c_a \sin r$
Therefore: $c_a \sin i = c_a \sin r$
and so the law of reflection is proved.

Refraction of light on corpuscular theory

- Newton assumed that there is an attraction between the molecules of a solid and the particles of light, and that this attraction acts only perpendicularly to the surface and only at very short distances from the surface.
- This has the effect of increasing the velocity of the light in the material



- Let the velocity of light in air be c_a and the velocity of light in the material in Figure be c_m .
- The velocity parallel to the material is unaltered and therefore:
$$c_a \sin i = c_m \sin r$$

Therefore:

$$c_m / c_a = \sin i / \sin r = {}_a n_m$$
- This ratio is the refractive index, but because $n > 1$ the velocity of light in the material must be greater than that in air.
- Newton accepted this result and other scientists preferred it to that of Huygens, mainly because of Newton's distinction

Main drawbacks of corpuscular theory

- This theory **could not** explain the phenomena of **interference, diffraction and polarization of light** etc
- According to this theory, the **velocity of light in denser medium is greater than the velocity of light in rarer medium** but this is proved wrong later
- This theory assumes that the **source of light loses the mass as it emits corpuscles**; but not such detriment in mass of the source of light is detected.
- This theory proposes that **velocity of the corpuscles increases as the temperature of the source increases**; experiments have proved that the velocity of light is independent of temperature.

- As light consists of small particles, emission of particles from a source causes reduction in the mass of source of light. It has been **experimentally proved** that there is **no reduction in the mass of the source of light**.
- If light is incident on a transparent material like glass or water, some part of light is reflected and some part of light is refracted. This simultaneous phenomenon of **reflection and refraction cannot be explained on the basis of corpuscular theory**
- According to Newton different colors of light are due to the different sizes and masses of the particles. But this fact **was not experimentally proved**
- The application of Newton's theory to refraction concludes that velocity of light in air medium is less than velocity of light in denser medium but **experimentally it was found to be wrong**
- Several phenomena of light such as **double refraction, interference, diffraction and polarization of light cannot be explained** on the basis of corpuscular theory.

The Wave Theory

- Discovered by **Christian Huygens**, a Dutch scientist, also in the seventeenth century
- States that light is emitted in a series of waves that spread out from a light source in all directions.
- These waves are not affected by gravity.
- Furthermore, **he disagreed with Newton** and said that light traveling from air to water will *decrease* the speed, and vice versa.
- Huygens was proved later to be correct
- 100 years later, Englishman **Thomas Young** completely **disproved the corpuscular theory** by showing that light waves can interfere with each other.



In 1678, Dutch physicist, *Christiaan Huygens*, believed that light was made up of waves vibrating up and down perpendicular to the direction of the light travels, and therefore formulated a way of visualizing wave propagation. This became known as '*Huygens' Principle*'.

Wave theory of Light

1. According to Huygen, Light is a **form of energy** , light is **propagated in the form of waves**.
2. The waves are emitted from a source of light and **travel** with a very **high velocity**.
3. The waves can **propagate** in all **possible directions** with a **constant velocity** in a homogeneous medium.
4. When these waves enter our eyes, we get the **sensation of sight**.
5. The **different colors** of light are due to the **different wavelengths** of the light waves.
6. Light waves have very **short wave length**
7. To explain propagation of light through vacuum, Huygen assumed that there exist a **hypothetical medium called as ether** (the whole space is filled with an imaginary medium called Ether)
8. **Light can propagate through ether** i.e. Each point in a source of light sends out waves in all directions in hypothetical medium called "ETHER".

Huygen's wave theory- *Features*

- The locus (a set of all points) of all medium particles vibrating in the same phase is called a **wave front**.
- This theory assumes that the **light waves are mechanical and transverse in nature**.
- This theory **is successful in explaining** the phenomena of **reflection, refraction, interference and diffraction** phenomena of light.
- This theory also assumes in similar to the previous theory that the **velocity of light is dependent on the density of the medium but color depends on the wavelength of the light**.

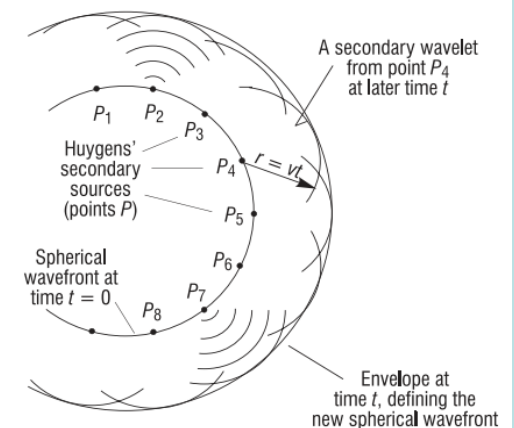
Huygens's Principle

- Huygens' principle gives a geometrical construction for finding the position of a wavefront at a future instant if its position is known at some particular instant
- Huygen's construction is based on the following two fundamental postulates:

Huygen's Principle

1. Every particle of the medium situated on the wavefront acts as a new wave-source from which fresh waves originate. These waves are called “secondary wavelets”
 - The secondary wavelets are spherical and travel in the medium in all directions with the same frequency and velocity as original wave
2. The surface, which touches all the wavelets from the secondary sources, gives the new position of wavefront

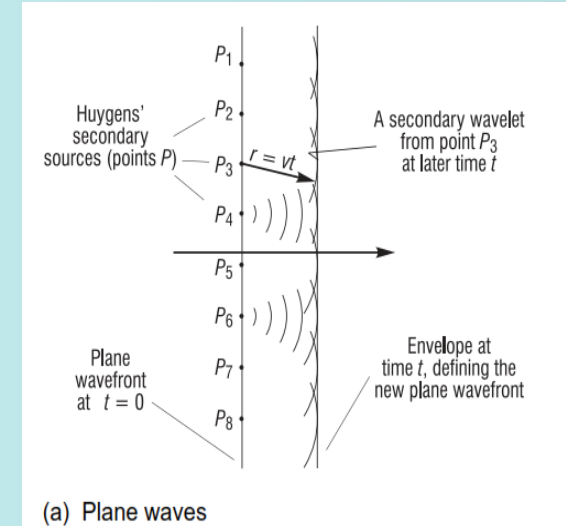
“The locus of all the points of the medium to which all the waves will reach simultaneously, so that all the points are in the same state of vibration is called wavefront”



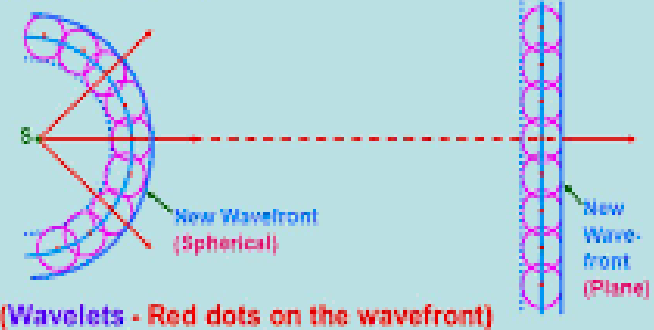
(b) Spherical waves

Huygen's construction of a plane wavefront

- Consider a point source of light placed at a point in vacuum or in air medium.
- The wavefront originating from the point source of light at very large distance is plane wavefront.
- According to Huygen's principle each and every point of the wavefront acts as secondary source of light and emits secondary waves in all possible direction with the same velocity in the same medium.
- Let 'v' be the velocity of light in air medium.
- To determine the position of new wavefront after time 't' draw spheres assuming each and every point of wavefront as its centre and vt as its radius.
- The tangential surface which touches all spheres is the position of new plane wavefront after time t.
- From figure it appears that there is a back wavefront but the waves moving in backward direction does not exist



Huygens' Construction or Huygens' Principle of Secondary Wavelets:

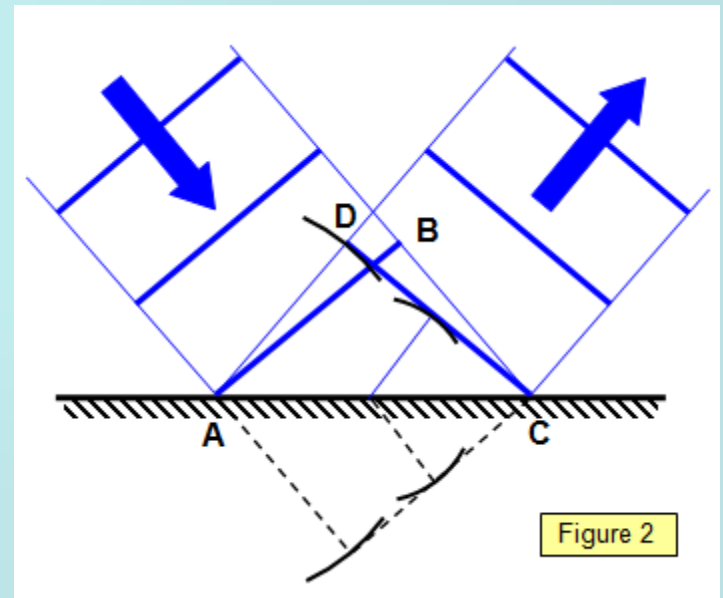


Note: Backward wavefront is rejected. Why?

Amplitude of secondary wavelet is proportional to $\frac{1}{2}(1+\cos\theta)$. Obviously, for the backward wavelet $\theta = 180^\circ$ and $(1+\cos\theta)$ is 0.

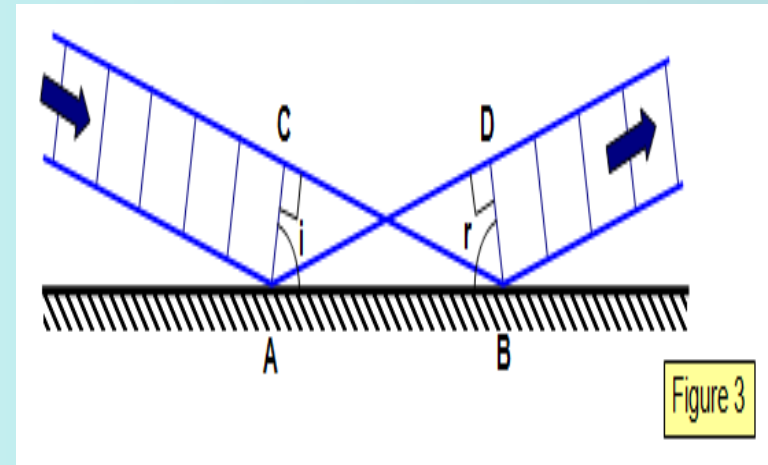
Reflection of a plane wavefront by a plane mirror

- The reflection of a plane wavefront by a plane mirror is shown
- Notice the initial position of the wavefront (AB), the secondary wavelets and the final position of the wavefront (CD).
- Notice that the shape of the wavefront is not affected by reflection at a plane surface
- The lines below the mirror show the position that the wavefront would have reached if the mirror had not been there.



Reflection

- Consider a parallel beam of monochromatic light incident on a plane surface, as shown in Figure
- The wave fronts will be plane both before and after reflection, since a plane surface does not alter the shape of waves falling on it.



Consider a point where the wavefront AC has just touched the mirror at edge A. While the light travels from A to D, that from C travels to B. The new envelope for the wavefront AC will be BD after reflection.

Therefore: $AD = CB$, Angle $ACB = \text{angle } ADB = 90^\circ$, AB is common

Therefore $\triangle ACB$ and $\triangle BDA$ are similar and so angle $CAB = \text{angle } BAD$.
Therefore $i = r$ and the law of reflection is proved.

Refraction

- Consider a plane monochromatic wave hitting the surface of a transparent material of refractive index n . The velocity of light in the material is c_m and that in air c_a . Now in Figure 4,

$$CB = AB \sin i$$

$$AD = AB \sin r$$

The same argument applies about the new envelope as in the case of reflection:

$$\text{time to travel } CB = CB/c_a = AB \sin i/c_a$$

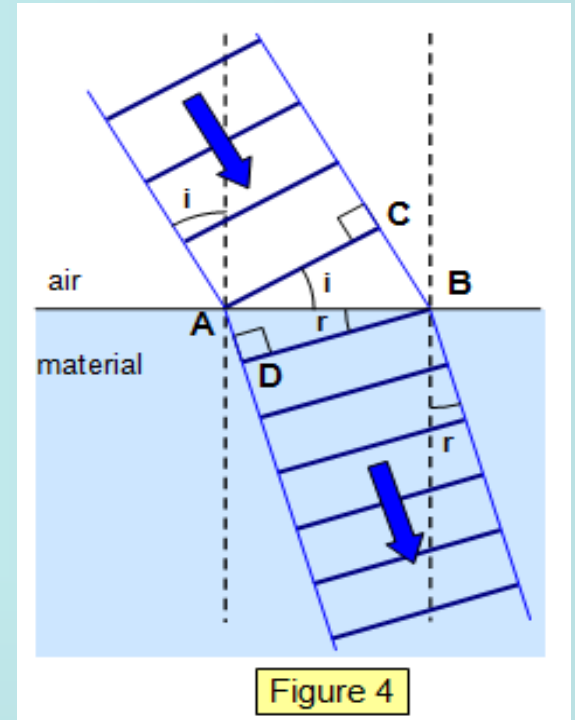
$$\text{time to travel } AD = AD/c_m = AB \sin r/c_m$$

But these are equal and therefore:

$$c_a/c_m = \sin i/\sin r = {}_a n_m.$$

This is Snell's law, and it was verified later by Foucault and others.

Notice that since the refractive index of a transparent material is greater than 1, Huygens' theory requires that the velocity of light in air should be greater than that in the material.



Merit's of Huygen's wave theory

1. On the basis of wave theory, phenomenon **of reflection and refraction can be satisfactorily explained.**
2. The application of wave theory to refraction concludes that the **velocity of light in a rarer medium is greater than the velocity of light in a denser medium.**
3. Several phenomena of light such as **double refraction, interference diffraction of light can be satisfactorily explained on the basis of wave theory.**

Demerit's of Huygen's wave theory

1. **Rectilinear propagation of light** cannot be explained on the basis of the wave theory of light.
2. It **fails** to explain **polarization, photoelectric effect and Compton effect**
3. For the propagation of light through vacuum, Huygen assumed presence of ether medium. **There is no existence of a medium 'ether'.**
4. It **fails** to explain **why light has no longitudinal component**

Maxwell's Electromagnetic Theory of Light

- The Scottish physicist James Clerk **Maxwell** postulated electromagnetic theory in the nineteenth century
- He contributed a great deal to the field of physics during the 1860s, with his **theories** on magnetism, optics and electricity completely transforming the realm of scientific discovery.

At the end of the nineteenth century, James Clerk Maxwell combined electricity, magnetism and light into one theory. He called his theory the electromagnetic theory.



James Clerk
Maxwell
(13 June 1831 –
5 November
1879)

According to Maxwell, light was an electromagnetic wave. Because light was an electromagnetic wave, light carried the same properties as the other electromagnetic waves. Maxwell was able to predict the speed of light by using electrical and magnetic constants. His calculations proved to be very close to the accepted value.

ELECTROMAGNETIC THEORY OF LIGHT

- Light waves possess **electrical and magnetic properties** and can **travel through a vacuum**. (i.e. light waves do not require a medium for transmission)
- Light eventually was proved to be **electromagnetic**. Light waves are a part of a larger family of electromagnetic waves and make up the electromagnetic spectrum.
- Current light theory says that light is made up of very small packets of electromagnetic energy called photons (the smallest unit of radiant energy).
- These photons move at a constant speed in the medium through which they travel.
- Photons move at a faster speed through a vacuum than they do in the atmosphere, and at a slower speed through water than air.

Maxwell's electro magnetic wave theory -*Features*

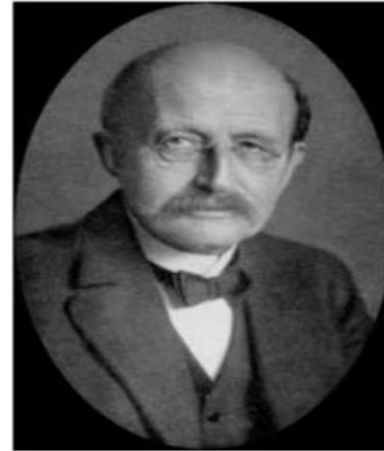
- The electric and magnetic fields in a electro magnetic wave are continuously varying with respect to time and space.
- At any instant electric and magnetic fields are perpendicular to each other and also perpendicular to the direction of light.
- The electro magnetic wave is a transverse wave.
- At every point in the wave at a given instant of time the electric and magnetic field strengths are equal.
- The velocity of propagation of electro magnetic wave depends on the electric and magnetic properties of the medium.

Drawbacks

- The main drawback of this theory is, it failed to explain the photo electric effect and Compton effect.
- Electro magnetic theory also failed in explaining the black body radiation

Planck's Quantum theory of light

- Quantum theory was put forward by Max-Planck in 1905. According to quantum theory “Energy radiated or absorbed can not have any fractional value. This energy must be an integral multiple of a fixed quantity of energy.



Max Karl Ernst Ludwig Planck, (April 23, 1858 – October 4, 1947) was a German physicist who discovered quantum physics, initiating a revolution in natural science and philosophy. He is regarded as the founder of quantum theory, for which he received the Nobel Prize in Physics in 1918

This quantity is called “QUANTUM” OR Energy released or absorbed is always in the form of packets of energy or bundles of energy. These packets of energy are known as QUANTA or PHOTONS

Later, it was proved that the correct and most accurate theory was the Quantum Theory.

Planck's Quantum theory

- According to this theory, light energy is released from source discretely in the form of energy packets of specific frequencies called photons or quanta.
- Photons are propagated as waves and if necessary interact with matter as particles.
- This phenomena could explain Compton effect and black body radiation.
- He formulated the photon theory of light and explained photoelectric effect

Drawbacks

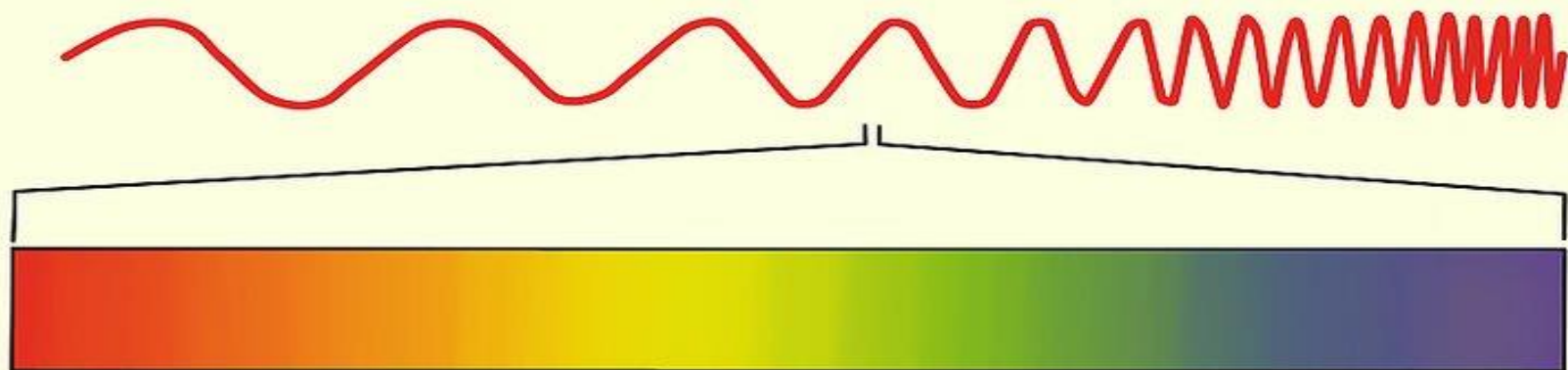
- Though the quantum theory explains successfully the interaction of radiation with matter, it cannot account for the phenomenon of polarization, interference and diffraction
- The main drawback of this theory is, it could not explain how it is connected with wave nature of light (dual nature).

The Dual Nature of Light

- A complete description of the nature of light requires **two explanations.**
- Light travels in the form of waves. The simplest demonstration of the wave nature of light is to throw two stones into a pool of water
- Light travels in the form of bundles or packets of energy called photons. This is sometimes called the particle nature of light. Light will be reflected from a solid mirror surface in the same way that a ball will bounce off a wall.

- Light exhibits dual nature
- All of the different behaviors of light can be explained by one (or both) of its wave and particle properties.
- **The reason for the wave-particle dualism may be understood** as follows:
- At the lower frequency end of the electromagnetic spectrum are radio waves whose wavelengths are so large that they spread over a very large volume of space
- Therefore, the energy available at any point is insignificantly small and their particle nature cannot be observed

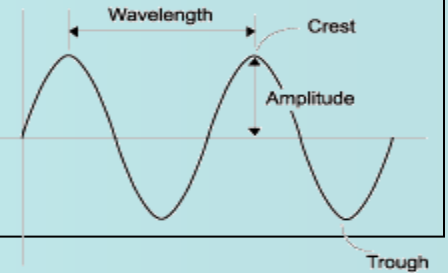
- At the higher frequency side of the electromagnetic spectrum are x and gamma rays whose wavelengths are so short that the wave energy is concentrated in a point of very small dimension
- The particle nature is readily experienced in this situation whereas the wave nature is less noticeable
- The visible region represents the transition region in which both the aspects of light can be observed



Descriptions of light

- Three descriptions namely **rays**, **waves** and **photons** appropriately describe the behavior of light
- When light is **reflected or refracted** by optical elements such as lenses, **ray** description is used,
- when **light propagates through space or any medium**, electromagnetic **wave** description is used
- Whenever **light interacts with matter** , **photon** description is used

Light as waves



- A wave is a regularly repeating disturbance
- The amplitude is the height of a wave.
- The wavelength is the distance between two successive crests or troughs in the wave. This can be measured in metres (m), micrometres (microns, $\mu\text{m} = 10^{-6} \text{ m}$), nanometres ($\text{nm} = 10^{-9} \text{ m}$) or angstroms ($\text{\AA} = 10^{-10} \text{ m}$).
- The frequency is the number of waves that pass by in 1 second. It has the unit Hertz (Hz).
- The velocity (speed) of a wave can be related to its frequency (, called 'Nu') and wavelength (, called 'Lambda') by the rule:

$$c = \nu \times \lambda$$

- where c = the speed of the light wave = $3 \times 10^8 \text{ ms}^{-1}$ (in a vacuum)

Sample problem:

Calculate the wavelength of red light in a vacuum if its frequency is 4×10^{14} Hz.

Solution

$$c = \nu \times \lambda$$

$$\lambda = \frac{c}{\nu} = \frac{3 \times 10^8}{4 \times 10^{14}} = 7.5 \times 10^{-7} \text{ m}$$

Light as Photons

- A photon is a small packet or bundle of energy.
- The energy of a photon is directly proportional to frequency (energy increases with increasing frequency) and is given by the rule: $E = h\nu$

- where h = Planck's constant = 6.63×10^{-34} Js
- Combining this rule with the rule for wave speed we obtain: $E = \frac{hc}{\lambda}$

Photon

- Photon is a particle that has no mass & no charge.
- Rest mass of photon is zero
- Photons are the carriers of electromagnetic radiation moving at the speed of light
- Photons have very high frequency.
- Energy of photon is $E=h\gamma$. Where h = Planck's constant ($h = 6.625 \times 10^{-34}$ J.s) and γ = Frequency of radiation
- A photon can be assumed as a discrete quantum of energy. Photons can be destroyed (or created) when radiation is absorbed or emitted.

Sample problem:

Calculate the energy of a 2.6 Å X-ray photon (remember that Å = 10^{-10} m).

Solution

$$E = \frac{hc}{\lambda} = \frac{(6.63 \times 10^{-34}) \times (3 \times 10^8)}{2.6 \times 10^{-10}} = 7.65 \times 10^{-16} \text{ Joule}$$

Note that the energy of X-rays is usually expressed in electron volts (eV):

$$1 \text{ J} = 6.24 \times 10^{18} \text{ eV}$$

Therefore, the energy of the 2.6 Å X-ray photon is equal to:

$$(7.65 \times 10^{-16}) \times (6.24 \times 10^{18}) = 4\,774 \text{ eV}$$

THANK YOU